

A $1.5 + \varepsilon$ Approximation Algorithm for Weighted Hypergraph Embedding in a Cycle *

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Abstract

The problem of Weighted Hypergraph Embedding in a Cycle (WHEC) is known to be NP-complete even when each hyperedge is unweighted or each weighted hyperedge contains exactly two vertices. In this paper, we propose an approximation algorithm for the WHEC problem to provide a solution with approximation bound of $1.5(opt + w_{max})$, where opt represents the optimal value of the problem and w_{max} denotes the largest weight of hyperedges. For any fixed $\varepsilon > 0$, we also present a polynomial-time approximation scheme to provide an embedding whose congestion is at most $(1.5 + \varepsilon)$ times the optimum. This improves previous results for the general WHEC problem.

1 Introduction

The problem of Weighted Hypergraph Embedding in a Cycle (WHEC) is to embed m weighted hyperedges of a hypergraph as paths in an n -vertex cycle, such that the maximum congestion of any physical link in the cycle is minimized. The WHEC problem is NP-complete even when each hyperedge is unweighted or each weighted hyperedge contains exactly two vertices [1].

In a special case when each hyperedge is unweighted, the unweighted version of the WHEC problem has many applications in electronic design automation [2, 3]. Several studies related to

the unweighted WHEC problem have been performed. Frank et al. [4] and Gonzalez et al. [2] proposed linear time algorithms to solve the problem when all hyperedges contain exactly only two vertices. Canley and Cohoon [5] showed that the unweighted WHEC problem is NP-complete in general, but solvable in polynomial time if the maximum congestion is bounded by a constant. Gonzalez [6] proposed two improved approximation algorithms that both generate solutions with maximum congestion at most two times the optimum. Carpenter et al. [7] provided a linear time approximation algorithm which routes the hyperedges in the clockwise direction starting from the lowest numbered vertex to the highest numbered vertex. This algorithm is also guaranteed to find a solution whose value is at most twice the optimal value. Recently, Gu and Wang presented an algorithm to solve the unweighted WHEC problem with the performance ratio 1.8 by a re-embedding technique [8]. Lee and Ho presented an LP-based algorithm to solve the problem with the performance ratio 1.5 [9, 10]. Deng and Li proposed a polynomial time approximation scheme (PTAS) for the unweighted WHEC problem by a randomized rounding approach [11].

In the other special case when each weighted hyperedge contains exactly two vertices, the graph version of the WHEC problem is equivalent to the ring loading problem without demand splitting. When each demand must be entirely routed in either of the two directions, the ring loading problem is NP-complete [12]. Cosares and Saniee presented a polynomial time algorithm that approximates the optimal solution value to within

*This work was supported in part by Taiwan NSC under grant no. NSC 94-2213-E-274-004.

a multiplicative factor of two [12]. Schrijver, Seymour and Winkler developed an efficient algorithm to generate a solution that exceeds the optimum by at most an additive term of 1.5 times the maximum weight [13]. Khanna proposed a polynomial time approximation scheme that computes a solution with at most $(1 + \varepsilon)$ times the optimum for any fixed $\varepsilon > 0$ [14].

The WHEC problem has many applications, including minimizing communication congestions in computer networks and parallel computations. These applications focus on minimizing the maximum congestions of multicast transmissions over any physical communication link. For example, in computer networks, the hypergraph embedding problem is equivalent to a multicast congestion problem [17, 18, 19] when each hyperedge is formed as a multicast tree (i.e., Steiner tree). Several studies related to the multicast congestion problem on more general graphs have been performed. Vempala and Vöcking [17] presented a randomized rounding algorithm for approximating multicast congestion to within $O(\log n)$ times the optimum. Carr and Vempala [18] proposed a randomized asymptotic algorithm for the multicast congestion problem to find a solution of congestion $O(\text{opt} + \log n)$, where opt is the optimal value of the maximum congestion. Jansen and Zhang [19] presented an improved approximation algorithm to overcome the difficulties of an exponential number of variables.

We are concerned with the general WHEC problem. In [1], an LP-based rounding algorithm and a linear time algorithm had been proposed to find a solution with maximum congestion at most two times the optimum. In this paper, we propose an improved $(2/3)$ -rounding algorithm to provide a solution with approximation bound of $1.5(\text{opt} + w_{\max})$, where opt represents the optimal value of the problem and $w_{\max}$ denotes the largest weight of hyperedges. For any fixed $\varepsilon > 0$, we also present a polynomial-time approximation scheme to provide an embedding whose congestion is at most $(1.5 + \varepsilon)$ times the optimum. This improves the previous approximation results [1] for the general WHEC problem.

2 Notation and Problem Definition

We assume that a cycle consists of n vertices labelled clockwise by 1 through n and each edge in the cycle is referred as an undirected link. The cycle can be denoted as $C = (V, E_c)$, where

$V = \{1, 2, \dots, n\}$ is a set of vertices and $E_c = \{(i, i+1) | 1 \leq i \leq n-1\} \cup \{(n, 1)\}$ is a set of undirected links. A hypergraph $H = (V, E_h)$ with m weighted hyperedges is defined over the same set of vertices V , where $E_h = \{h_1, h_2, \dots, h_m\}$ is a set of hyperedges. The hyperedge h_i consists of $|h_i|$ vertices with a non-negative weight w_i for interconnecting these vertices. In particular, these interconnected vertices of hyperedge h_i are represented as an ordered sequence $(v_1^i, v_2^i, \dots, v_{|h_i|}^i)$, i.e. $v_1^i \leq v_2^i \leq \dots \leq v_{|h_i|}^i$.

We denote the j -th adjacent path of hyperedge h_i as $p(i, j)$, where $p(i, j)$ is a clockwise connecting path between v_j^i and v_{j+1}^i in the cycle C , and the last adjacent path $p(i, |h_i|)$ of the hyperedge h_i is connected between $v_{|h_i|}^i$ and v_1^i . Obviously, vertices $v_1^i, v_2^i, \dots, v_{|h_i|}^i$ in the hyperedge h_i must be connected by at least $|h_i| - 1$ adjacent paths to ensure the connectivity. We define an assignment of adjacent paths to embed the hypergraph H in the cycle C as a set of binary variables $Y = \{y_{p(i,j)} | \forall i \in \{1, 2, \dots, m\}, j \in \{1, 2, \dots, |h_i|\}\}$, where $y_{p(i,j)} = 1$ if an adjacent path $p(i, j)$ is embedded in the cycle C , and $y_{p(i,j)} = 0$ otherwise. Therefore a feasible assignment of adjacent paths can be expressed as $\sum_{1 \leq j \leq |h_i|} y_{p(i,j)} \geq |h_i| - 1$,

$\forall h_i \in E_h$. Moreover, we denote $P(e)$ as a set of adjacent paths that pass through the link $e \in E_c$. Therefore, the WHEC problem can be formally defined as follows:

Definition 1 *The problem of minimizing the Maximum Congestion for Weighted Hypergraph Embedding in a Cycle (WHEC): Given a cycle $C = (V, E_c)$ and a weighted hypergraph $H = (V, E_h)$, find a feasible assignment Y of adjacent paths to embed the hypergraph H in the cycle C such that the maximum congestion φ over any link in E_c is minimized. Note that the maximum link congestion of an assignment Y can be expressed as*

$$\varphi = \max_{e \in E_c} \left\{ \sum_{p(i,j) \in P(e)} y_{p(i,j)} w_i \right\}, \text{ where } y_{p(i,j)} = 1 \text{ if the } p(i, j) \text{ is embedded in the cycle } C, \text{ otherwise } y_{p(i,j)} = 0.$$

3 LP-based approximation

The WHEC problem can be modelled as an Integer Linear Programming (ILP) Formulation. An assignment of adjacent paths to embed the hypergraph H in the cycle C can be represented by a set of binary variables $Y =$

$\{y_{p(i,j)} \mid \forall i \in \{1, 2, \dots, m\}, j \in \{1, 2, \dots, |h_i|\}\}$, where $y_{p(i,j)} = 1$ if an adjacent path $p(i, j)$ is embedded in the cycle C , and $y_{p(i,j)} = 0$ otherwise. Our objective is to minimize the maximum congestion φ over all links in E_c . We formulate an ILP model to solve the problem as follows:

$\Phi(Y)$: Minimize φ

Subject to

$$\sum_{1 \leq j \leq |h_i|} y_{p(i,j)} \geq |h_i| - 1, \quad \forall h_i \in E_h, i \in \{1, 2, \dots, m\} \quad (\text{connectivity constraints})$$

$$\sum_{p(i,j) \in P(e)} y_{p(i,j)} w_i \leq \varphi, \quad \forall e \in E_c \quad (\text{capacity constraints})$$

$$y_{p(i,j)} \in \{0, 1\}, \quad \forall i \in \{1, 2, \dots, m\}, j \in \{1, 2, \dots, |h_i|\} \quad (\text{binary variables})$$

There are two classes of constraints in this problem. First, the connectivity constraints mean that each hyperedge $h_i \in E_h$ must be connected by at least $|h_i| - 1$ adjacent paths. Secondly, the capacity constraints ensure that each physical link $e \in E_c$ can only be embedded at most the maximum congestion φ of hyperedges. Finally, the objective function $\Phi(Y)$ of the ILP is to minimize the maximum congestion φ .

Suppose that we relax the integer constraints of Y and require only that $0 \leq y_{p(i,j)} \leq 1, \forall i, j$; this linear program is called the LP relaxation of the ILP formulation, and the LP solution immediately provides a lower bound on the minimum congestion. On the other hand, finding a tight upper bound for the ILP formulation presents a difficult challenge. However, the LP-rounding technique can commonly be applied to find an upper bound and generate an approximate solution. Let $\varphi^* = \Phi(Y^*)$ and $\varphi^L = \Phi(Y^L)$ be the optimal solutions of ILP formulation and LP relaxation respectively. If the solution for the LP-rounding is φ^R , then we have $\varphi^L \leq \varphi^* \leq \varphi^R$.

The next lemma states that the value of the k -th smallest variable in $\{y_{p(i,j)}^L \mid 1 \leq j \leq |h_i|\}$ is at least $(k-1)/k$ for every hyperedge $h_i \in E_h$. This implies that the value of the second smallest variable is at least $1/2$. Therefore, a $(1/2)$ -rounding approach can be applied to generate an approximate solution $\varphi^R = \Phi(Y^R)$, where we assign $y_{p(i,j)}^R = 1$ if $y_{p(i,j)}^L \geq 1/2$, and $y_{p(i,j)}^R = 0$ otherwise. The $(1/2)$ -rounding approach always produce a 2-approximated solution for the WHEC problem [1].

Lemma 1 *For every hyperedge $h_i \in E_h$, $|h_i| \geq k$, $i \in \{1, 2, \dots, m\}$, the value of the k -th smallest*

variable in $\{y_{p(i,j)}^L \mid 1 \leq j \leq |h_i|\}$ is at least $(k-1)/k$.

4 1.5($opt + w_{max}$) approximation

In this section, we apply a $(2/3)$ -rounding approach to generate an approximate solution $\varphi^R = \Phi(Y^R)$, where we assign $y_{p(i,j)}^R = 1$ if $y_{p(i,j)}^L \geq 2/3$, and $y_{p(i,j)}^R = 0$ otherwise. Despite the simplicity of the $(2/3)$ -rounding approach, Lemma 2 states that the maximum congestion of the rounding approach is at most 1.5 times the optimal congestion of the WHEC problem when the value of the smallest LP variable in $\{y_{p(i,j)}^L \mid 1 \leq j \leq |h_i|\}$ is at most $1/3$, i.e. $\min_{1 \leq j \leq |h_i|} \{y_{p(i,j)}^L\} \leq 1/3$ for every hyperedge $h_i \in E_h, i \in \{1, 2, \dots, m\}$.

Lemma 2 *When $\min_{1 \leq j \leq |h_i|} \{y_{p(i,j)}^L\} \leq 1/3$ for every hyperedge $h_i \in E_h, i \in \{1, 2, \dots, m\}$, the maximum congestion of the $(2/3)$ -rounding approach is at most 1.5 times the optimum.*

Proof: Let $\varphi^R = \Phi(Y^R)$ be the solution of the $(2/3)$ -rounding approach. First, we show that $\Phi(Y^R)$ is a feasible solution to the problem if the condition of $\min_{1 \leq j \leq |h_i|} \{y_{p(i,j)}^L\} \leq 1/3, \forall h_i \in E_h$ is satisfied. According to connectivity constraints, any solution to LP relaxation must satisfy that $\sum_{1 \leq j \leq |h_i|} y_{p(i,j)}^L \geq |h_i| - 1$ for every hyperedge $h_i \in E_h, i \in \{1, 2, \dots, m\}$. If the value of the smallest LP variable in $\{y_{p(i,j)}^L \mid 1 \leq j \leq |h_i|\}$ is at most $1/3$, we know that the value of the second smallest LP variable is at least $2/3$. This implies that the sum of rounded values $\sum_{1 \leq j \leq |h_i|} y_{p(i,j)}^R$ is at least $|h_i| - 1$ for all $i \in \{1, 2, \dots, m\}$. Each hyperedge $h_i \in E_h$ is continuously connected, hence the solution $\Phi(Y^R)$ is feasible. Next, we show that $\varphi^R \leq 1.5\varphi^*$. If $y_{p(i,j)}^R$ is rounded to 1, then $y_{p(i,j)}^L$ has a value at least $2/3$. Therefore, we have $y_{p(i,j)}^R \leq (3/2)y_{p(i,j)}^L, \forall i, j$. From the capacity constraints, we have $\sum_{p(i,j) \in P(e)} y_{p(i,j)}^R w_i \leq \varphi^R \leq \sum_{p(i,j) \in P(e)} (3/2)y_{p(i,j)}^L w_i \leq (3/2)\varphi^L \leq (3/2)\varphi^*, \forall e \in E_c$. $\square$

When $\min_{1 \leq j \leq |h_i|} \{y_{p(i,j)}^L\} \leq 1/3, \forall h_i \in E_h$, the $(2/3)$ -rounding approach generates a feasible solution whose value is at most 1.5 times the optimum.

However, the solution of the (2/3)-rounding algorithm may be infeasible if the condition is unsatisfied. Therefore, we will propose a $1.5(opt + w_{max})$ approximation algorithm on the condition that $\min_{1 \leq j \leq |h_i|} \{y_{p(i,j)}^L\} > 1/3, \forall h_i \in E_h$.

Let us focus on the smallest three variables in $\{y_{p(i,j)}^L \mid 1 \leq j \leq |h_i|\}$ for each hyperedge $h_i \in E_h$, $i \in \{1, 2, \dots, m\}$. Let $y_{g(i,1)}^L$, $y_{g(i,2)}^L$ and $y_{g(i,3)}^L$ respectively be the first, second and third smallest LP variables for the hyperedge h_i . Note that $g(i,j)$ denotes the adjacent path with the j -th smallest LP variable for the hyperedge h_i . From Lemma 1, the value of the third smallest variable $y_{g(i,3)}^L$ is at least $2/3$. If the (2/3)-rounding algorithm is applied to generate an approximate solution, the third smallest variable and the $|h_i| - 3$ larger variables will certainly be rounded to 1. If the condition $\min_{1 \leq j \leq |h_i|} \{y_{p(i,j)}^L\} \leq 1/3$ is unsatisfied, values of the first and second smallest variables are both less than $2/3$ and will be truncated to 0 by the (2/3)-rounding algorithm. In this case, the hyperedge h_i is disconnected and we need to round either the first or second smallest variable to 1 for ensuring the connectivity of the hyperedge.

Let D be the set of disconnected hyperedges in the output of the (2/3)-rounding algorithm, i.e., $D = \left\{ h_i \mid \sum_{1 \leq j \leq |h_i|} y_{p(i,j)}^R = |h_i| - 2, 1 \leq i \leq m \right\}$. For each disconnected hyperedge $h_i \in D$, we assume that the adjacent path $g(i, 2)$ for the second smallest variable of h_i is connected around the ring from vertex s_i to vertex t_i in the clockwise direction and the adjacent path $g(i, 1)$ is connected in the other way around the ring from t_i to s_i .

The problem of rounding selection for the smallest two variables on $h_i \in D$ can be translated into an unsplitting subproblem of the ring loading problem. For each disconnected hyperedge $h_i \in D$, we define $y_{g(i,2)}^M = \frac{3}{2}y_{g(i,2)}^L$ and $y_{g(i,1)}^M = 1 - y_{g(i,2)}^M$ respectively. If the adjacent path $g(i, 2)$ for the second smallest LP variable $y_{g(i,2)}^L$ of h_i is connected around the ring from vertex s_i to vertex t_i in the clockwise direction, then we assume that the adjacent path $g(i, 1)$ is connected from vertex t_i to vertex s_i in the other way around the ring. We need to round either $y_{g(i,1)}^M$ or $y_{g(i,2)}^M$ to 1 for ensuring the connectivity of the hyperedge h_i . Therefore, our objective is to find an optimal rounding assignment of $y_{g(i,1)}^R$ and $y_{g(i,2)}^R$ for each disconnected hyperedge $h_i \in D$ such that the maximum increment of load, denoted as Δ , over any physical link in the ring is minimized. The next

lemma shows that the translation is valid.

Lemma 3 *Let D be the set of disconnected hyperedges in the output of a (2/3)-rounding algorithm, i.e., $D = \left\{ h_i \mid \sum_{1 \leq j \leq |h_i|} y_{p(i,j)}^R = |h_i| - 2, 1 \leq i \leq m \right\}$. If we define $y_{g(i,2)}^M = \frac{3}{2}y_{g(i,2)}^L$ and $y_{g(i,1)}^M = 1 - y_{g(i,2)}^M$ for each disconnected hyperedge $h_i \in D$, then we have $y_{g(i,1)}^M \leq \frac{3}{2}y_{g(i,1)}^L - y_{g(i,1)}^R$, $y_{g(i,2)}^M \leq \frac{3}{2}y_{g(i,2)}^L - y_{g(i,2)}^R$ and $y_{g(i,1)}^M \leq \frac{3}{2}y_{g(i,j)}^L - y_{g(i,j)}^R, 3 \leq j \leq |h_i|$.*

Proof: Consider a disconnected hyperedge $h_i \in D$ from the output of (2/3)-rounding algorithm. From Lemma 1, the value of the third smallest LP variable $y_{g(i,3)}^L$ of h_i is at least $2/3$. Since the hyperedge h_i is disconnected, both the first and second smallest LP variables should be less than $2/3$, i.e., $y_{g(i,1)}^L \leq y_{g(i,2)}^L < \frac{2}{3}$. Therefore, we have $y_{g(i,1)}^R = 0$, $y_{g(i,2)}^R = 0$ and $y_{g(i,j)}^R = 1, \forall j \geq 3$. It is trivial that $y_{g(i,2)}^M \leq \frac{3}{2}y_{g(i,2)}^L - y_{g(i,2)}^R$, since we define $y_{g(i,2)}^M = \frac{3}{2}y_{g(i,2)}^L$ and we know that $y_{g(i,2)}^R = 0$. Next we show that $y_{g(i,1)}^M \leq \frac{3}{2}y_{g(i,1)}^L - y_{g(i,1)}^R$. From the definition, we have $y_{g(i,1)}^M = 1 - \frac{3}{2}y_{g(i,2)}^L$. Since $y_{g(i,1)}^R = 0$ and we know that $y_{g(i,1)}^L + y_{g(i,2)}^L \geq 1$ from Lemma 1, we have $y_{g(i,1)}^M = 1 - \frac{3}{2}y_{g(i,2)}^L \leq 1 - \frac{3}{2}(1 - y_{g(i,1)}^L) \leq \frac{3}{2}y_{g(i,1)}^L - \frac{1}{2} < \frac{3}{2}y_{g(i,1)}^L - y_{g(i,1)}^R$. Finally, we show that $y_{g(i,1)}^M \leq \frac{3}{2}y_{g(i,j)}^L - y_{g(i,j)}^R, 3 \leq j \leq |h_i|$. Since $y_{g(i,1)}^R = 0$, $y_{g(i,1)}^L < \frac{2}{3}$ and we know that $y_{g(i,1)}^L + y_{g(i,2)}^L + y_{g(i,3)}^L \geq 2$ from Lemma 1, we have $y_{g(i,2)}^L \geq 2 - y_{g(i,1)}^L - y_{g(i,3)}^L > 2 - \frac{2}{3} - y_{g(i,3)}^L = \frac{4}{3} - y_{g(i,3)}^L$. Then, for $j \geq 3$, we have $y_{g(i,2)}^L \geq \frac{4}{3} - y_{g(i,j)}^L$ since $y_{g(i,j)}^L \geq y_{g(i,3)}^L$. We thus have $y_{g(i,1)}^M = 1 - \frac{3}{2}y_{g(i,2)}^L \leq 1 - \frac{3}{2}(\frac{4}{3} - y_{g(i,j)}^L) = \frac{3}{2}y_{g(i,j)}^L - 1 = \frac{3}{2}y_{g(i,j)}^L - y_{g(i,j)}^R$, for any $j \geq 3$. $\square$

Next, we will extend the merging and sequential rounding techniques [13, 15] from the ring loading problem for ensuring $\sum_{p(i,j) \in P(e)} y_{p(i,j)}^R w_i \leq \frac{3}{2}(w_{max} + \sum_{p(i,j) \in P(e)} y_{p(i,j)}^L w_i) \leq \frac{3}{2}(opt + w_{max})$ for all $e \in E_c$. Here, opt represents the optimal value of the problem and w_{max} denotes the largest weight of hyperedges, i.e., $opt = \varphi^*$ and $w_{max} = \max\{w_i \mid 1 \leq i \leq m\}$. In order to apply the merging and sequential rounding techniques for the unsplitting problem, we represent the notations of adjacent paths $g(i, 1)$ and $g(i, 2)$ for each disconnected hyperedge $h_i \in D$ in the clockwise order around the ring. Let $s(i, 1)$ and $s(i, 2)$ respectively be the first and second adjacent paths

of h_i in the clockwise order around the ring (sequencing from vertex 1). Now, given the adjacent paths $s(i, 1)$ and $s(i, 2)$ with translative variables $y_{s(i,1)}^M$ and $y_{s(i,2)}^M$ for all disconnected hyperedge $h_i \in D$, our objective is to find an optimal rounding assignment of $y_{s(i,1)}^R$ and $y_{s(i,2)}^R$ such that the maximum increment of load, denoted as Δ , over any physical link in the ring is minimized. Two disconnected hyperedges h_i and h_j are said to be *crossing* if both their two adjacent paths $s(i, 1)$, $s(i, 2)$ and $s(j, 1)$, $s(j, 2)$ intersect on at least one physical link. Otherwise, they are said to be *non-crossing*. The next lemma describes how to merge two non-crossing hyperedges into a connected hyperedge without increasing the total load over any physical link.

Lemma 4 *If two disconnected hyperedges are non-crossing, they can be merged into a connected hyperedge without increasing the total load over any physical link.*

Proof: Suppose that two disconnected hyperedges h_i and h_j are non-crossing. Without loss of generality, we assume that their non-crossing adjacent paths are given as Fig. 1(a). In this figure, we assume that $a = w_i$, $a_1 = y_{s(i,1)}^M w_i$, $a_2 = y_{s(i,2)}^M w_i$, $b = w_j$, $b_1 = y_{s(j,1)}^M w_j$ and $b_2 = y_{s(j,2)}^M w_j$ respectively. If $y_{s(j,2)}^M w_j \geq (1 - y_{s(i,1)}^M) w_i$ (see Fig. 1(b)), we output an unsplitting assignment to ensure the connectivity of the hyperedge h_i by setting $y_{s(i,1)}^R = y_{s(i,1)}^M = 1$, $y_{s(i,2)}^R = y_{s(i,2)}^M = 0$ and $D = D - \{h_i\}$. Then we define a new splitting assignment for the hyperedge h_j by setting $y_{s(j,1)}^M = y_{s(j,1)}^M + y_{s(i,2)}^M (w_i/w_j)$ and $y_{s(j,2)}^M = y_{s(j,2)}^M - (1 - y_{s(i,1)}^M) (w_i/w_j)$. The new load on each physical link is either unchanged or reduced. And, the new splitting assignment for the hyperedge h_j keeps the same property of $y_{s(j,1)}^M + y_{s(j,2)}^M = 1$. On the other hand, if $y_{s(j,2)}^M w_j < (1 - y_{s(i,1)}^M) w_i$ (see Fig. 1(c)), we output an unsplitting assignment to ensure the connectivity of hyperedge h_j by setting $y_{s(j,1)}^R = y_{s(j,1)}^M = 1$, $y_{s(j,2)}^R = y_{s(j,2)}^M = 0$ and $D = D - \{h_j\}$. Similarly, we define a new splitting assignment for the hyperedge h_i by setting $y_{s(i,1)}^M = y_{s(i,1)}^M + y_{s(j,2)}^M (w_j/w_i)$ and $y_{s(i,2)}^M = y_{s(i,2)}^M - (1 - y_{s(j,1)}^M) (w_j/w_i)$. In this case, the load on each physical link and the property for the hyperedge h_i are also properly maintained. $\square$

The merging procedure can be performed repeatedly without increasing the maximum load

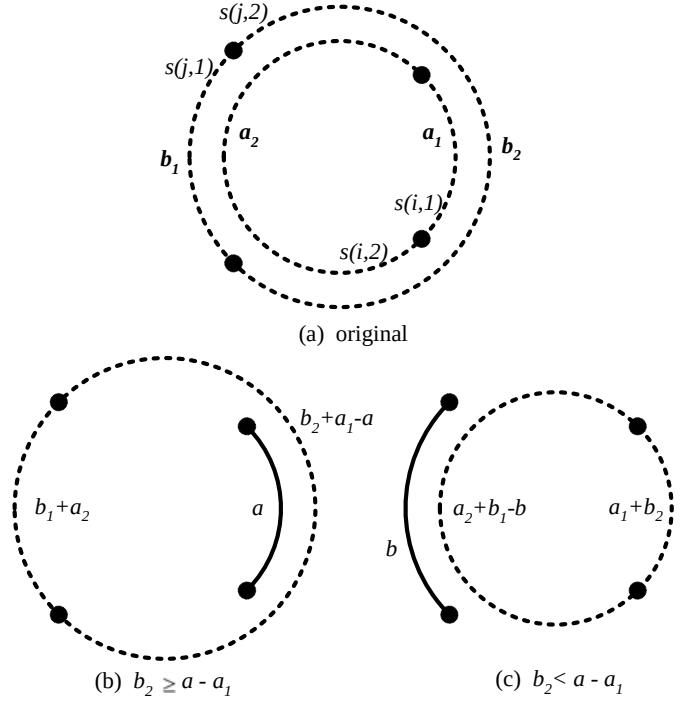


Figure 1: Merge two disconnected hyperedges into a connected hyperedge.

over any physical link, i.e., the maximum load increment Δ in the ring is zero. Since each time we reduce by one the total number of disconnected hyperedges, this procedure will terminate in at most $|D|$ steps until no two disconnected hyperedges contain a pair of non-crossing adjacent paths. Nevertheless, not all adjacent paths of disconnected hyperedges are non-crossing. Let S be the set of remnant disconnected hyperedges. We perform a sequential rounding technique to round each remnant variable in the clockwise direction around the ring. In the following algorithm, we combine the (2/3)-rounding, merging and sequential rounding techniques for ensuring an approximation bound of $1.5(opt + w_{max})$. Here, opt represents the optimal value of the problem and w_{max} denotes the largest weight of hyperedges, i.e., $w_{max} = \max\{w_i \mid 1 \leq i \leq m\}$.

Theorem 1 *The improved (2/3)-rounding algorithm has an approximation bound of $1.5(opt + w_{max})$ where opt represents the optimal value of the problem and w_{max} denotes the largest weight of hyperedges.*

Proof: Step (6) ensures that $(y_{s(r_1,1)}^R - y_{s(r_1,1)}^M) w_{r_1} \in [-\frac{w_{r_1}}{2}, \frac{w_{r_1}}{2})$ and $(y_{s(r_1,2)}^R -$

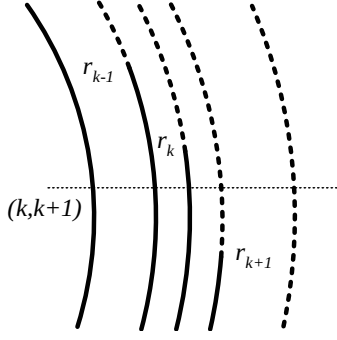


Figure 2: The link $(k, k+1)$ is crossed by the adjacent paths $s(r_1, 1), s(r_2, 1), \dots, s(r_k, 1), s(r_{k+1}, 2), s(r_{k+2}, 2), \dots, s(r_{|S|}, 2)$.

$y_{s(r_1, 2)}^M)w_{r_1} \in (-\frac{w_{r_1}}{2}, \frac{w_{r_1}}{2}]$. By induction, if the partial sum $\sum_{i \in \{r_1, \dots, r_{k-1}\}} (y_{s(i, 1)}^R - y_{s(i, 1)}^M)w_i$ from r_1 to r_{k-1} are given and places in the interval $[-\frac{w_{max}}{2}, \frac{w_{max}}{2}]$, then we define $y_{s(k, 1)}^R$ sequentially around the ring from r_1 to $r_{|S|}$ by step (7) for ensuring $\sum_{i \in \{r_1, \dots, r_k\}} (y_{s(i, 1)}^R - y_{s(i, 1)}^M)w_i \in [-\frac{w_{max}}{2}, \frac{w_{max}}{2}]$. Due to the complementary nature, we also have $\sum_{i \in \{r_1, \dots, r_k\}} (y_{s(i, 2)}^R - y_{s(i, 2)}^M)w_i \in (-\frac{w_{max}}{2}, \frac{w_{max}}{2}]$.

For any physical link $(k, k+1) \in E_c$, we assume without loss of generality that it is crossed by two partitions of sequential hyperedges $r_1, r_2, \dots, r_k$ and $r_{k+1}, r_{k+2}, \dots, r_{|S|}$, as shown in Fig. 2. Therefore the increment of congestion of the link $(k, k+1)$ is at most $\Delta_k = \sum_{i \in \{r_1, \dots, r_k\}} (y_{s(i, 1)}^R - y_{s(i, 1)}^M)w_i + \sum_{i \in \{r_{k+1}, \dots, r_{|S|}\}} (y_{s(i, 2)}^R - y_{s(i, 2)}^M)w_i = \sum_{i \in \{r_1, \dots, r_k\}} (y_{s(i, 1)}^R - y_{s(i, 1)}^M)w_i + \sum_{i \in \{r_1, \dots, r_{|S|}\}} (y_{s(i, 2)}^R - y_{s(i, 2)}^M)w_i - \sum_{i \in \{r_1, \dots, r_k\}} (y_{s(i, 2)}^R - y_{s(i, 2)}^M)w_i \leq \frac{w_{max}}{2} + \frac{w_{max}}{2} - (-\frac{w_{max}}{2}) = \frac{3}{2}w_{max}$. We thus have $\Delta = \max_k \{\Delta_k\} \leq \frac{3}{2}w_{max}$. Therefore, from Lemma 2, we have an approximation bound of $1.5(opt + w_{max})$. $\square$

5 $(1.5 + \varepsilon)$ approximation

The improved $(2/3)$ -rounding algorithm generate a feasible solution whose value is at most $1.5(opt + w_{max})$. Clearly, if the optimal value opt of maximum congestion is large, the additional term

of the maximum weight w_{max} of hyperedges can be translated into a relatively small deviation in value from the optimal solution. However, the value of optimal solution may be near to the maximum weight of hyperedges, and the additive error term could be a significant deviation from the optimum. In this section we will apply the polynomial time approximation scheme (PTAS) in [14] to show that for any fixed $\varepsilon > 0$, there is a polynomial time algorithm that yields a $(1.5 + \varepsilon)$ -Approximation.

Let φ^T denote the value of approximate solution of the $(1/2)$ -rounding algorithm [1]. The approximate solution of the $(1/2)$ -rounding algorithm is always feasible and it ensures that $\varphi^T \leq 2\varphi^*$. On the other hand, if the improved $(2/3)$ -rounding algorithm is applied to generate an approximate solution, we need to round either the first or second smallest variable of a disconnected hyperedge to 1 for ensuring the connectivity. We denote S as the set of these disconnected hyperedges. A hyperedge in S is said to be *heavy* if its weight is at least $\frac{1}{3}\varepsilon\varphi^T$, and *light* otherwise. Let S_H and S_L respectively denote the subsets of heavy and light hyperedges in S . The following lemma bounds the total number of heavy hyperedges in S_H .

Lemma 5 *The set S_H contains at most $\frac{3n}{2\varepsilon}$ hyperedges, i.e., $|S_H| \leq \frac{3n}{2\varepsilon}$.*

Proof: Each hyperedge in S_H has weight at least $\frac{1}{3}\varepsilon\varphi^T \leq \frac{2}{3}\varepsilon\varphi^*$. In an ideal case, these hyperedges are routed in one hop and shared out equally among n physical links in the ring. Obviously, the maximum congestion on any physical link is at least $\frac{2}{3}\varepsilon\varphi^* \frac{|S_H|}{n}$. This gives a lower bound on the maximum congestion for embedding all hyperedges in S_H . Hence we have $\frac{2}{3}\varepsilon\varphi^* \frac{|S_H|}{n} \leq \varphi^*$. Rearranging the term, we get $|S_H| \leq \frac{3n}{2\varepsilon}$. $\square$

For every disconnected hyperedge in S_H , we need to choose one of two adjacent paths for ensuring the connectivity. If the longer of the two adjacent paths is selected for connecting a disconnected hyperedge along the ring, we say that the hyperedge is routed in the long way. The next lemma bounds the total number of heavy hyperedges that could be routed in the long way for any optimal routing.

Lemma 6 *In any optimal routing, at most $3/\varepsilon$ heavy hyperedges are embedded in the long way.*

Proof: Suppose to the contrary that more than $3/\varepsilon$ heavy hyperedges were routed in the long way in the optimal solution. Since the hyperedge

must be routed at least $\lceil n/2 \rceil$ physical links in the long way, the total weight induced by those heavy hyperedges in the long way is more than $\lceil n/2 \rceil \times \frac{2}{3}\varepsilon\varphi^* \times 3/\varepsilon \geq n\varphi^*$. By pigeonhole principle, some links must have congestion more than the optimal value $\frac{n\varphi^*}{n} = \varphi^*$, which contradicts the optimality of the optimal routing. $\square$

Since the largest weight of hyperedges in S_L is less than $\frac{1}{3}\varepsilon\varphi^T \leq \frac{2}{3}\varepsilon\varphi^*$, we can perform the clockwise sequential rounding algorithm on S_L to generate an approximation with bound of $\frac{3}{2}w_{max} < \frac{3}{2} \times \frac{2}{3}\varepsilon\varphi^* = \varepsilon\varphi^*$. On the other hand, we need exhaustively try all possible ways to find the optimal solution for embedding all hyperedges in S_H . Fortunately, the time complexity of the exhaustive search is bounded by a polynomial function of the problem size.

Lemma 7 *For an exhaustive search on S_H , the total number of times for searching is bounded by $k(\frac{en}{2})^k$, where $k = 3/\varepsilon$.*

Proof: Since the set S_H contains at most $\frac{3n}{2\varepsilon}$ hyperedges, we need only to choose at most $3/\varepsilon$ heavy hyperedges from S_H for embedding them in the long way. Let $k = 3/\varepsilon$. The number of choices is bounded by $\sum_{i=0}^k \binom{\frac{kn}{2}}{i}$. From the well-known inequality $\binom{x}{y} \leq (\frac{ex}{y})^y$, we have

$$\sum_{i=0}^k \binom{\frac{kn}{2}}{i} \leq \sum_{i=0}^k \left(\frac{e \times kn}{2i} \right)^i \leq k \left(\frac{en}{2} \right)^k$$

Thus, the total running time of exhaustive search on S_H is bounded by a polynomial function of n . $\square$

Theorem 2 *For any fixed $\varepsilon > 0$, there is a polynomial-time algorithm whose approximate solution is within $(1.5 + \varepsilon)$ times the optimum for the general WHEC problem.*

6 Conclusion

This paper concerns with the problem of Weighted Hypergraph Embedding in a Cycle (WHEC) and focuses on the impact of the maximum congestion. We formulate the problem as an Integer Linear Program (ILP) and propose an improved $(2/3)$ -rounding algorithm to provide a solution with approximation bound of $1.5(opt + w_{max})$, where opt represents the optimal value of the WHEC problem and w_{max} denotes the largest

weight of hyperedges. If the value of optimal solution is near to the maximum weight of hyperedges, then the additive error term w_{max} could be a significant deviation from the optimum. For any fixed $\varepsilon > 0$, we also present a polynomial approximation algorithm to provide an embedding whose congestion is at most $(1.5 + \varepsilon)$ times the optimum. To our knowledge, this is the best approximation bound known for the general problem of embedding weighted hypergraph in a cycle. However, the main problem with the LP-based approximation is the time required to solve the linear program. To improve the efficiency, it is worthwhile to look into ways of making better use of the congestion information for obtaining a simple heuristic algorithm with tighter approximation bound.

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